

Coffee Space

Business Requirements Document · Portfolio / Pilot Baseline · Version 2.0

Café Operations Platform: POS · Inventory · Purchasing · Floor · Reports · Roles · Multi-branch Context

Document field	Value
Document type	Business Requirements Document (BRD)
Product	Coffee Space — Café Operations Platform
Version	2.0 — polished portfolio / pilot baseline
Role context	Business Analysis · Data Analysis · Business Development
Audience	Product stakeholders, café operators, POS/ERP teams, hiring managers reviewing BA work
Delivery assets	Working prototype, BRD, case-study presentation.

1. Executive Summary

Coffee Space is a browser-based working prototype for independent and small-chain cafés. It unifies point of sale, inventory, purchasing, floor tables, profitability reporting and role-based access in one operating workflow.

The product was defined from an operations-first perspective: every confirmed sale should immediately affect stock, cost of goods, and same-day performance, while cashier access remains limited to transaction execution.

2. Business Problem

- Sales often live in a POS while stock and purchases live in spreadsheets, notebooks or disconnected tools.
- Recipe cost and COGS are rarely tied to every ticket, so gross margin is estimated rather than measured.
- Cashiers may see cost and margin fields that should be restricted to managers.
- Multi-location cafés often lack a simple branch context on every transaction.
- Waste, expenses, shifts and cash movements are recorded inconsistently, weakening audit trails.

3. Desired State

- One workflow for sell → deduct stock → update cost → reflect P&L.
- Clear separation between Manager and Cashier capabilities.
- Branch-aware sales and operational records.
- Exportable operational data for Excel / Google Sheets analysis.
- A clean UX that makes the system usable for a cashier in under 30 seconds on the core sale path.

4. Goals and Success Metrics

Goal	Success indicator
Reduce time to know daily net performance	A confirmed sale appears in dashboard/reports within the same session.
Prevent unauthorized cost visibility	Cashier cannot open manager modules or see unit costs, inventory value, COGS or margins.
Keep inventory visible without a separate tool	Manager dashboard shows inventory value and low-stock alerts.
Support dine-in workflow	Dine-in orders require table selection; takeaway does not.
Support analysis outside the system	CSV export produces order-line data usable in Excel or Google Sheets.

5. Stakeholders and Access Needs

Role	Needs	System access
Café Manager	Full P&L, stock control, catalog control, users, reports	All modules
Cashier	Fast, accurate selling only	Limited dashboard + POS
Owner / Investor	Margin visibility, branch comparison, top products	Manager-level reports
Business Analyst	Process clarity, requirements, rules, acceptance criteria	Requirements owner / UAT

6. Scope

In scope v1	Out of scope v1
Authentication with Manager and Cashier roles	Hardware drivers: printers, cash drawers, barcode scanners
POS with takeaway / dine-in, payment method and table selection	Online ordering and delivery aggregator integrations
Prepared products with recipes and retail packaged SKUs	Loyalty programs and gift cards
Raw-material and retail stock, waste logging, inventory value	Full accounting ledger and tax filing
Purchases with weighted average cost and operating expenses	Native mobile apps
Dashboard, day/week/month reports and CSV export	Server-side database and multi-device sync, planned for next phase
Arabic / English UI, light/dark theme, branch context	

7. As-Is / To-Be Process Flow

Current fragmented flow	Coffee Space target flow
Sale recorded in POS only	Sale confirmed in POS and tagged to branch/user/table/payment
Stock checked manually later	Recipe materials or retail packs deduct immediately
COGS calculated manually or approximated	COGS calculated from recipe/pack cost × quantity sold
Expenses live separately	Operating expenses recorded and included in P&L
Daily performance only clear at month-end	Dashboard updates same-day sales, gross, expenses and net performance
Cashier may see sensitive fields	Role-based access hides cost and margin data from cashier

8. Functional Requirements

ID	Area	Requirement
FR-A1	Access & Session	Authenticate users by username and password.
FR-A2	Access & Session	Enforce Manager vs Cashier module visibility.
FR-A3	Access & Session	Bind each sale to current user and selected branch.
FR-P1	POS	Search and add active products to cart.
FR-P2	POS	Cart shows quantity, unit price, line total and grand total.
FR-P3	POS	Dine-in orders require active table selection.
FR-P4	POS	Confirmed sale deducts recipe materials or retail packs.
FR-P5	POS	Block sale when available stock is insufficient.
FR-I1	Inventory	Maintain raw material quantity, minimum and unit cost.
FR-I2	Inventory	Maintain retail pack quantity, minimum and pack cost.
FR-I3	Inventory	Flag low-stock items at or below minimum.
FR-I4	Inventory	Record waste with quantity, reason and value impact.
FR-B1	Purchasing	Maintain suppliers.
FR-B2	Purchasing	Purchase entry increases stock and recalculates weighted average cost.
FR-B3	Expenses	Record operating expenses by category and amount.
FR-C1	Products	Support categories and product CRUD.
FR-C2	Products	Prepared products store recipe material quantities.
FR-T1	Floor	Create/update/delete tables with seats and area.
FR-R1	Reporting	Dashboard shows sales, COGS-based gross, expenses, net, AOV, inventory value and alerts for manager.
FR-R2	Reporting	Reports support day/week/month periods and CSV export.

ID	Area	Requirement
FR-O1	Operations	Record customers, stock movements, transfers, cash entries, shifts and audit events.

9. Permissions Matrix

Module / Data	Manager	Cashier
Dashboard revenue	Yes	No
COGS / gross / net profit	Yes	No
Inventory value and unit costs	Yes	No
POS transactions	Yes	Yes
Dine-in table selection	Yes	Yes
Purchases and suppliers	Yes	No
Product pricing and recipes	Yes	No
Reports and CSV export	Yes	No
Users and permissions	Yes	No
Audit log	Yes	No

10. Key Data Entities

- User, Role, Branch, Category, Product, RecipeLine
- RawMaterial, RetailStock, Supplier, Purchase, Expense, Waste
- Table, Sale/Order, OrderLine, PaymentMethod, OrderType
- Customer, StockMovement, Transfer, CashEntry, Shift, AuditEvent

11. Business Rules

- BR1 — Only active products appear on POS.
- BR2 — Sale is rejected if any line exceeds available stock.
- BR3 — Dine-in without table is rejected.
- BR4 — Weighted average cost updates only on purchase of that item.
- BR5 — Cashier cannot access manager modules or cost fields.
- BR6 — Dashboard and reports respect the selected branch filter where implemented.
- BR7 — Reset/demo baseline does not fabricate sales, purchases or expenses.

12. Acceptance Criteria

AC	Acceptance criterion
AC1	Manager can complete a sale, see stock drop and see ticket on dashboard.
AC2	Cashier cannot open inventory, purchases, admin, reports, users or cost fields.
AC3	Purchase of a raw material changes quantity and weighted average unit cost correctly.

AC	Acceptance criterion
AC4	Waste reduces stock and is stored with reason.
AC5	Report period switch updates KPIs and CSV downloads order lines.
AC6	Dine-in forces table selection; takeaway does not.
AC7	Language and theme toggles update the interface.
AC8	Branding and navigation show Coffee Space only.

13. Requirements Traceability Matrix

Business goal	Functional requirements	Acceptance criteria
Daily performance visibility	FR-P4, FR-R1, FR-R2	AC1, AC5
Role-based operational control	FR-A1, FR-A2	AC2
Inventory visibility	FR-I1, FR-I2, FR-I3, FR-I4	AC1, AC3, AC4
Dine-in service support	FR-P3, FR-T1	AC6
External analysis	FR-R2	AC5

14. Future Phases

- Phase 2 — Server API, shared database, secure password hashing and multi-device sync.
- Phase 3 — Printer, cash drawer and barcode scanner integrations.
- Phase 4 — Promotions, loyalty, customer profiles and delivery integrations.
- Phase 5 — Forecasting, reorder suggestions, advanced analytics and multi-brand support.